

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI / ML engineer shipping production healthcare AI end to end: HIPAA-conscious RAG (~35% retrieval precision, >90% grounded alignment), predictive ML pipelines (15-20% recall on high-risk patient categories at >90% precision), agentic LLM workflows (>30% hallucination reduction), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Production computer vision via YOLOv8 (~30% inference latency reduction). Master's in Computer Science, Kent State. Targeting Aspen Skiing Company's AI Engineer seat - production RAG pipelines (~35% retrieval precision in regulated healthcare directly translate to enterprise data + LLM grounding for the Aspen One platform), Manga Lens 4-provider AI vision integration (proves multi-provider LLM API discipline transferable to Azure OpenAI + Anthropic Claude orchestration), Agentic Healthcare Claims with schema-validated JSON contracts (matches reusable AI infrastructure + workflow automation), Emerson SQL Server + T-SQL stored procedure tuning + Jenkins CI/CD (~20% query speedup, ~30% deploy-error reduction - direct Microsoft-stack credibility for Aspen's Azure-aligned platform), and FastAPI + Docker packaging with structured logging (matches monitoring + production lifecycle requirements). Open to Aspen / Denver CO relocation, F-1 OPT US-base.

CORE COMPETENCIES

Applied AI Engineer • RAG / Embeddings / Vector Search • LLM APIs (Azure OpenAI, Anthropic Claude) • LangChain / Agentic Workflows • Python (Production-grade) • SQL / Data Pipelines / ETL • REST APIs / CI/CD / Git • Microsoft Stack (SQL Server, T-SQL background) • Production AI Lifecycle / Monitoring • Aspen / Denver CO Relocation Open

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Applied AI, Healthcare

USA - Healthcare Technology, Applied AI

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision ~**35%**; retrieval-grounded response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; agent response stability ~**25%**; hallucinations >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement scoring, support prioritization; recall +**15-20%** on high-risk categories at >**90%** precision via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- EHR + appointment + ticket preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream instability ~**40%**.
- Authored **HIPAA-conscious data governance** and stakeholder-facing system-limitation docs; ran weekly stakeholder calls with clinical SMEs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; query execution time ~**20%**, data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; deployment errors >**30%**, release cycle ~**35-40%**.
- Designed **SQL Server DMVs and Grafana** performance dashboards for long-running queries, deadlocks, CPU/IO contention; incident recurrence ~**25%**, RCA speed ~**30%**.
- Implemented **RBAC and audit logging for financial modules** in a compliance-sensitive oil & gas environment.

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction cut manual review **60-70%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped, Chrome Web Store)

Real-time manga / webtoon translation extension shipped to the Chrome Web Store. **Multi-provider abstraction layer** integrating four AI vision providers (Claude Sonnet, GPT-4o-mini, GPT-4.1-Nano, Ollama/minicpm-v) with per-provider payload handling (WebP for cloud, JPEG for local), cost-aware routing, circuit-breaker fallback. Manifest V3 service workers; multi-section capture pipeline for tall panels with coordinate remapping; 7-day cache; per-domain selector configs for 29 manga/webtoon sites; narrowed host permissions and privacy policy.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama, multi-provider integration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake Normalization, Validation, Consistency Analysis, Duplicate Detection, Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector search, GPT-4, schema contracts

Dream Decoder - AI Multimodal Creative Intelligence

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar interpretation + Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** - **+30%** contextual alignment and **+25-30%** first-pass image success rate over naïve direct-prompt orchestration.

FastAPI, React, TypeScript, structured prompt schemas, multimodal orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with a unified **YOLOv8** detection-and-classification model - inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping + NMS tuning for stable real-time operation via OpenCV video capture.

PyTorch, YOLOv8, OpenCV, real-time inference

EDUCATION

Master of Science, Computer Science - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - KL University

Jun 2019 - May 2023

Founder / Lead of E-Farming platform (university project).

CERTIFICATIONS

Deep Learning Specialization - DeepLearning.AI (Coursera)

2024

SKILLS

LLM / GenAI: RAG, agentic workflows, structured outputs, evaluation pipelines, guardrails, grounding, vector search, LangChain, LlamaIndex, multi-provider LLM API (Anthropic Claude, OpenAI)

Languages: Python (production), TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++

ML Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers

Backend / Inference: FastAPI, Flask, REST, Docker, structured logging, microservices

Microsoft Stack: SQL Server, T-SQL, DMVs, stored procedure tuning, schema-versioned deploy via Jenkins

Data / ETL: Pandas, NumPy, ETL pipelines, data quality audits, time-series forecasting

DevOps: Docker, Jenkins, GitHub Actions, Grafana, CI/CD, cloud-ready deployment

Domains: Applied AI, Enterprise AI, Production-grade AI Lifecycle